THE QUANT DESK CHEATCODE

75 Short Tricks, Mnemonics & Mathematical Hacks
for Fast-Paced Quantitative Interviews

CORE SUBJECTS COVERED

Probability • Linear Algebra • Statistics • Stochastic Calculus
Option Pricing • Numerical Methods • Machine Learning
Brainteasers • Fixed Income • Model Validation

QUANTMENTOR

Expert Quant Mentorship & Premium Digital Resources

6.8 Trick 36: Game Theory – Dominant Strategies & Nash Equilibrium

★ Trick: Mixed Strategy Nash Equilibrium Shortcut

In a 2×2 zero-sum game with payoff matrix A:

$$\text{Row player mixes at} \quad p^* = \frac{a_{22} - a_{21}}{a_{11} - a_{12} - a_{21} + a_{22}},$$

$$\text{Column player mixes at} \quad q^* = \frac{a_{22} - a_{12}}{a_{11} - a_{12} - a_{21} + a_{22}}.$$

Minimax theorem: The game value $V = p^* a_{11}(1 - q^*) + \ldots$ (plug in p^*, q^*).

Dominant strategy shortcut: Eliminate dominated strategies first, then solve residual game.
Iterated elimination of dominated strategies (IEDS) always finds equilibrium if one exists via dominance.

♦ Mnemonic

"Mix to make opponent indifferent": Set p^* so column player's payoff is the same regardless of their choice. Then p^* must satisfy column-player's indifference condition.

Example: Matching Pennies

Payoff matrix: Heads/Heads $= +1$, Heads/Tails$= -1$, Tails/Heads$= -1$, Tails/Tails$= +1$.

$a_{11} = 1, a_{12} = -1, a_{21} = -1, a_{22} = 1$.
$p^* = (1 - (-1))/(1 - (-1) - (-1) + 1) = 2/4 = \mathbf{1/2}$. Mixed 50/50 for both. Game value $= 0$ (fair).

Example: Scissors-Paper-Rock – Why Uniform Mix

3×3 symmetric zero-sum game. By symmetry, each player mixes uniformly at $1/3$ each. Expected payoff $= 0$.

Any deviation (e.g., play Rock 50%) is immediately exploited by an opponent who shifts to Paper. Uniform mixing is the unique Nash equilibrium.

Example: Blotto Game Intuition

Allocate troops to 2 battlefields. You have 3 troops, opponent has 2. Optimal mixed strategy?

Deploy (2,1) or (1,2) with equal probability. Expected value > 0 since you have more troops. Pure strategies are always beaten.

Example: Pricing Game – Bertrand Equilibrium

Two firms price a homogeneous good, each firm's cost $= c$. Nash equilibrium price?

$P^* = c$ (Bertrand paradox). With 2 competitors and identical products, price is driven to marginal cost – zero economic profit even with just 2 firms.

Example: Auction Strategy – First Price

Two bidders, values $V_i \sim U[0, 1]$ i.i.d. Optimal bid in first-price sealed auction?

Optimal strategy (Bayes-Nash equilibrium): $b_i(v) = v/2$. Shade bid by 50%.
With n bidders: $b_i(v) = v(n-1)/n$. More competition $\Rightarrow$ bids closer to true value.

methodology should be used instead?

Financial data: Use **time-series CV** (train on past, validate on future) – never use standard k-fold which leaks future information (look-ahead bias).

Example: PCA for Dimensionality Reduction

A quantitative analyst has 200 financial features with only $T = 500$ daily observations. Assess the risk of overfitting, and explain how Principal Component Analysis (PCA) can be used to mitigate this issue.

$n/T = 200/500 = 0.4$. Very high risk. First apply PCA to reduce to $k \ll 200$ factors. Rule of thumb: keep components explaining $\geq 85\%$ of variance. Then run regression on the k principal components.

Example: Random Forest vs Boosting

Contrast Random Forest and Gradient Boosting (XGBoost) models in terms of their training processes (parallel vs. sequential) and their sensitivity to overfitting. Under what quantitative finance scenarios would you select one over the other?

In quant finance: Use RF for robustness (noisy data, many irrelevant features). Use XGBoost when you have cleaner signals and can tune carefully. Both benefit from rolling window or purged cross-validation.

10.2 Trick 61: Sharpe Ratio Manipulation & Statistics

★ Trick: Sharpe Ratio – Statistical Properties

$$\mathrm{SR} = \frac{\mu_r}{\sigma_r}$$

where μ_r and σ_r are excess return mean and std.

Annualised SR from daily: $\mathrm{SR}_{\mathrm{ann}} = \mathrm{SR}_{\mathrm{daily}} \times \sqrt{252}$.

Standard error: $\mathrm{SE}(\widehat{\mathrm{SR}}) \approx \sqrt{\frac{1+\frac{1}{2}\widehat{\mathrm{SR}}^2}{T}}$ (for i.i.d. returns, T years).

Non-i.i.d. correction: If returns are autocorrelated with lag-1 AC ρ_1: $\mathrm{SR}_{\mathrm{corrected}} \approx \mathrm{SR} \times \sqrt{\frac{1-\rho_1}{1+\rho_1}}$.

Deflated SR (DSR): Adjusts for multiple testing over n strategies: $\mathrm{DSR} = \Phi\left(\frac{\widehat{\mathrm{SR}} - \mathcal{E}[\max \widehat{\mathrm{SR}}]}{\mathrm{SE}(\widehat{\mathrm{SR}})}\right)$.

♦ Mnemonic

"Multiply daily SR by $\sqrt{252}$ for annual".
"SE $\approx \sqrt{(1 + SR^2/2)/T}$": to be significant, need $SR \times \sqrt{T} > 1.96$.

Example: SR Significance Test

A trading strategy has a daily Sharpe Ratio of 0.05. Calculate the annualized Sharpe Ratio. Over a 3-year history ($T = 3$), is this Sharpe Ratio statistically significant at the 5% level? How many years of historical data are needed to achieve significance?

t-stat $= \widehat{SR}\sqrt{T} = 0.79 \times \sqrt{3} = 1.37 < 1.96$. **Not significant** at 5%.
Need $T \geq (1.96/0.79)^2 = 6.15$ years for significance.

11.3 Drill 3: Stochastic Calculus Speed Round

Example: S3.1 – Ito's Lemma

$dX = dt + dW$. Find $d(X^2)$.

$d(X^2) = (2X + 1)dt + 2X\,dW$.

Example: S3.2 – GBM Log

$dS = 0.1S\,dt + 0.2S\,dW$. $d(\ln S)$?

$d(\ln S) = (0.1 - 0.02)dt + 0.2\,dW = \mathbf{0.08\,dt + 0.2\,dW}$.

Example: S3.3 – Martingale Check

Is $M_t = W_t^2 - t$ a martingale?

$dM = 2W\,dW + dt - dt = 2W\,dW$. No dt term: Yes, it is a martingale.

Example: S3.4 – Stochastic Integral Variance

$\text{Var}\left(\int_0^3 2\,dW_t\right)$?

$= \int_0^3 4\,dt = \mathbf{12}$ (Ito isometry).

Example: S3.5 – Expected Stock Price

$S_0 = 100$, $\mu = 8\%$, $\sigma = 20\%$, $T = 2$. $\mathbb{E}[S_2]$?

$\mathbb{E}[S_T] = S_0 e^{\mu T} = 100e^{0.16} = \mathbf{117.4}$.

Example: S3.6 – Quadratic Variation

$[\sigma W]_T$?

$= \sigma^2 T$.

Example: S3.7 – Feynman-Kac

$\frac{\partial u}{\partial t} + \frac{1}{2}\frac{\partial^2 u}{\partial x^2} = 0$, $u(x, T) = x^2$. $u(x, 0)$?

$u(x, 0) = \mathbb{E}[W_T^2 | W_0 = x] = \mathbb{E}[(x + W_T - W_0)^2] = x^2 + T = \mathbf{x^2 + T}$.

Example: S3.8 – Risk-Neutral Drift

$dS = 0.15S\,dt + 0.3S\,dW$, $r = 0.05$. Risk-neutral SDE?

$dS = 0.05S\,dt + 0.3S\,d\tilde{W}$.